Translation Lookaside Buffer (TLB)

To make one TLB entry, we need to put all the right values in these 4 registers

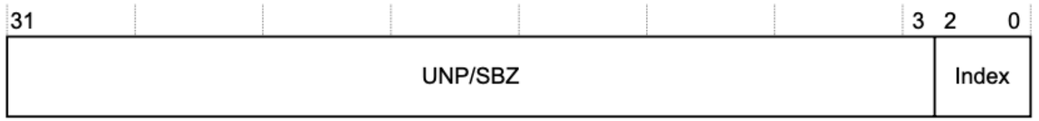

Figure 3-76 TLB Lockdown Index Register format

Figure 3-77 TLB Lockdown VA Register format

Figure 3-78 TLB Lockdown PA Register format

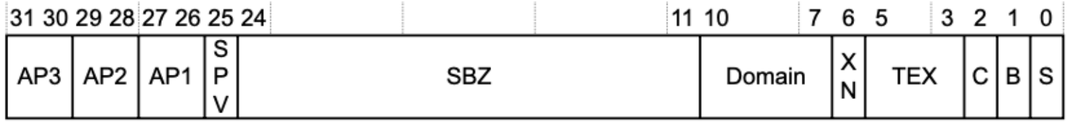

Figure 3-79 TLB Lockdown Attributes Register format

Lets try to understand this one by one!

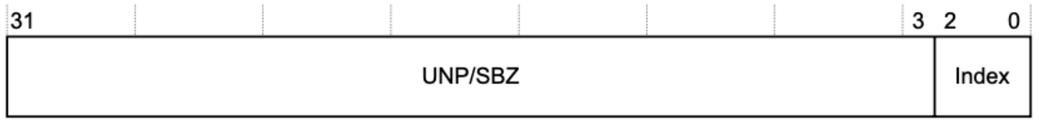

Figure 3-76 TLB Lockdown Index Register format

Figure 3-77 TLB Lockdown VA Register format

Figure 3-78 TLB Lockdown PA Register format

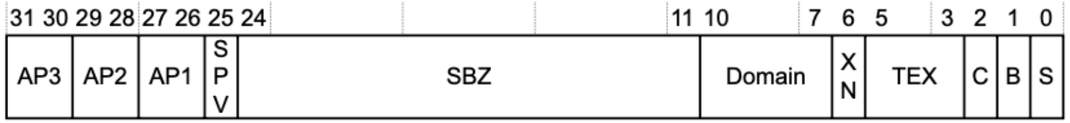

Figure 3-79 TLB Lockdown Attributes Register format

The first register is the **INDEX REGISTER**

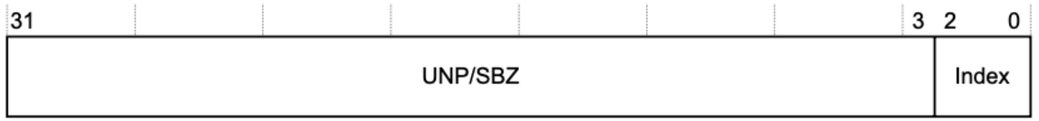

Figure 3-76 TLB Lockdown Index Register format

Figure 3-77 TLB Lockdown VA Register format

Figure 3-78 TLB Lockdown PA Register format

Figure 3-79 TLB Lockdown Attributes Register format

SBZ just means it should be zero

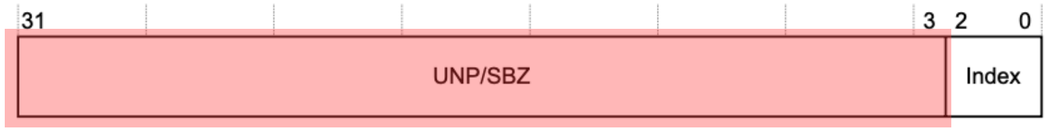

Figure 3-76 TLB Lockdown Index Register format

Bits	Field name	Function
[31:3]	-	UNP/SBZ.
[2:0]	Index	Selects the lockdown entry of the eight TLB lockdown entries to read or write when accessing other TLB lockdown access registers. Select lockdown entry 0 to 7.

This is the other section: **Index**

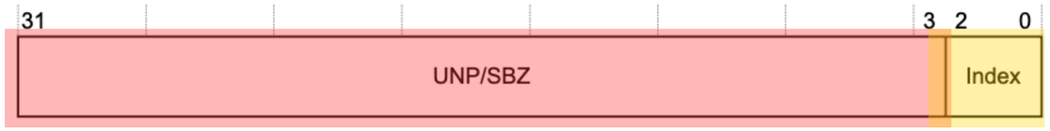

Figure 3-76 TLB Lockdown Index Register format

Bits	Field name	Function
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Lets look at how it works

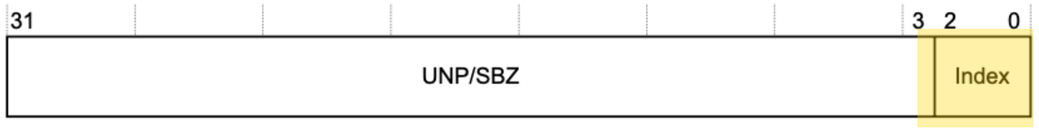

Figure 3-76 TLB Lockdown Index Register format

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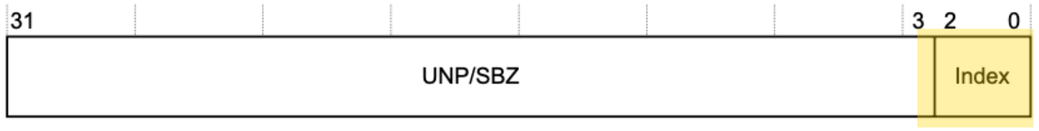
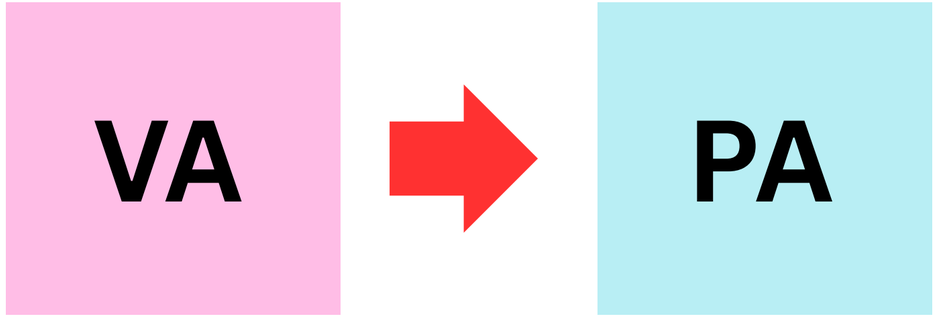


Figure 3-76 TLB Lockdown Index Register format

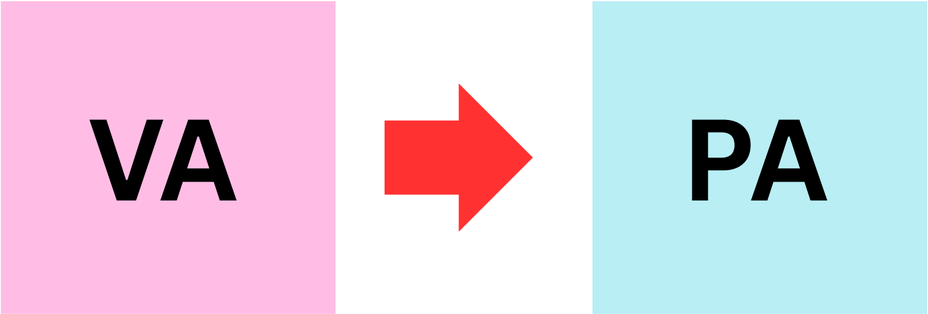
Bits	Field name	Function
[31:3]	-	UNP/SBZ.
[2:0]	Index	Selects the lockdown entry of the eight TLB lockdown entries to read or write when accessing other TLB lockdown access registers. Select lockdown entry 0 to 7.

Whenever we make a virtual address to physical address, we can store that mapping in one of the 8 TLB lockdown access regsiters

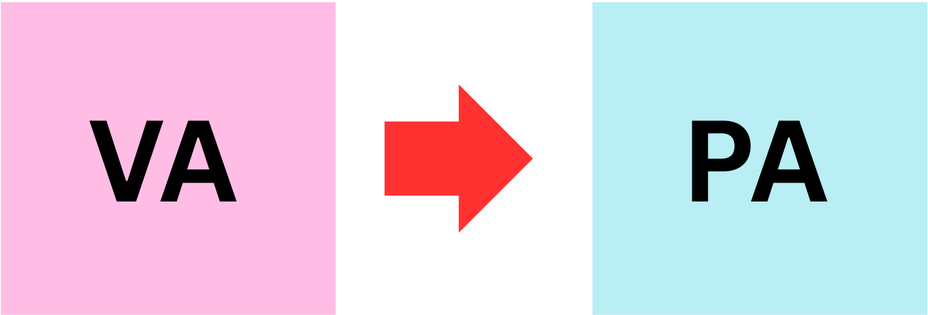
At a high level ...



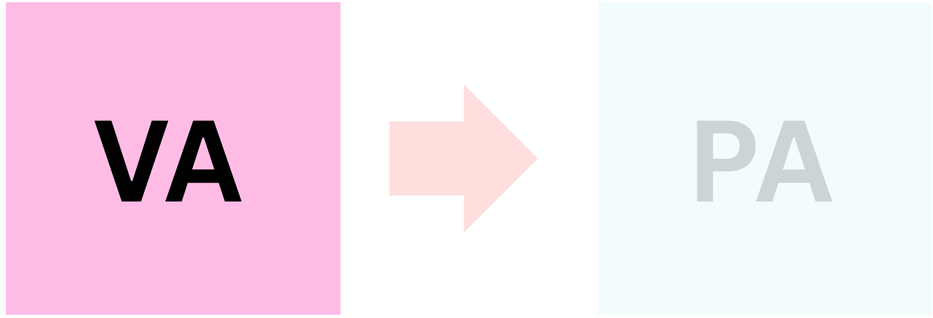
Virtual memory maps a Virtual Address to a Physical Address



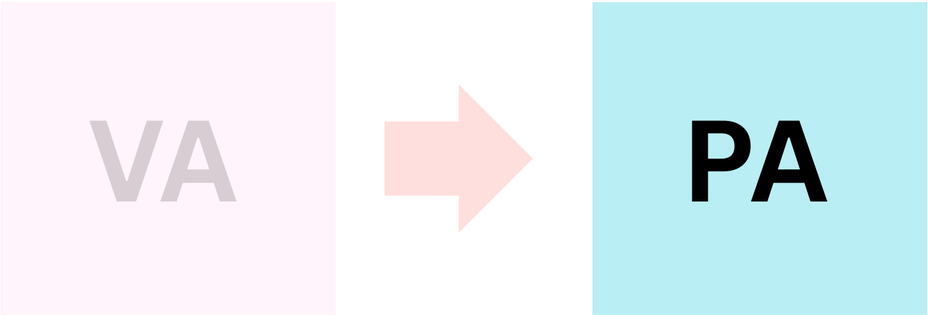
We need to write these mappings down somewhere



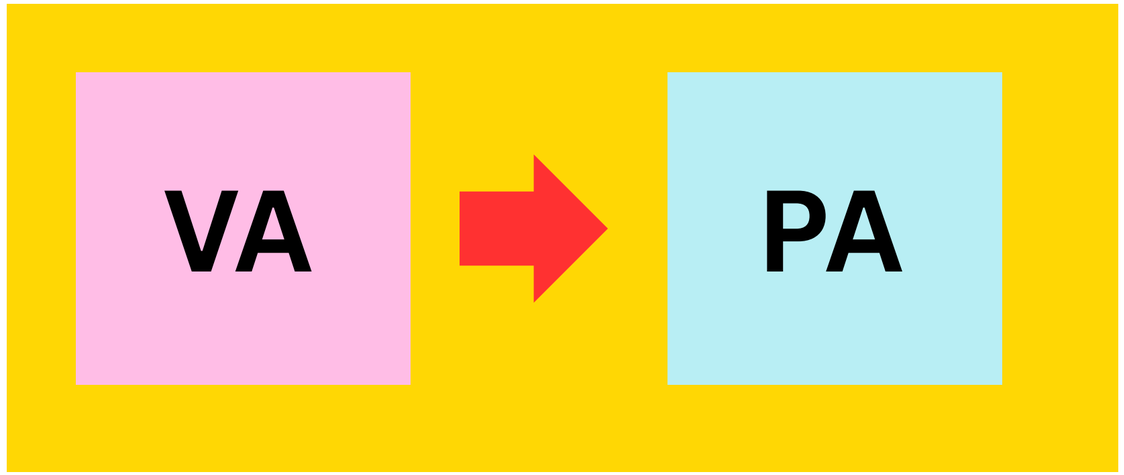
So that when we access a Virtual address



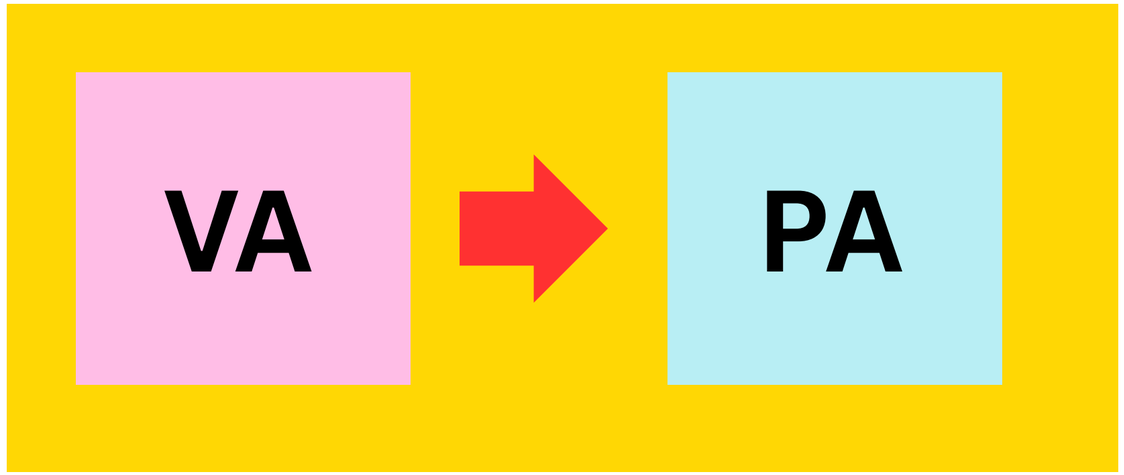
We know what Physical address we need to convert it to



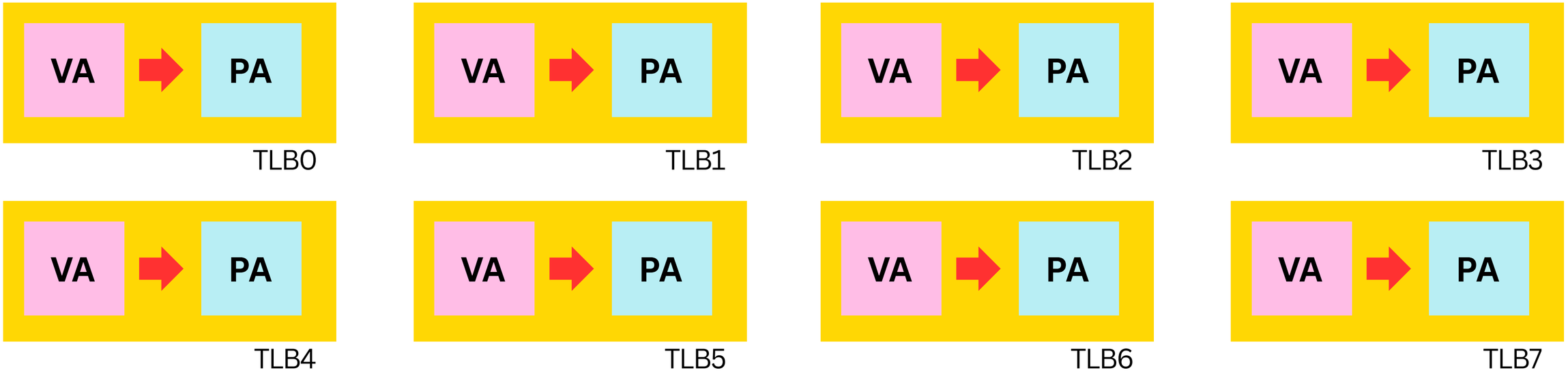
A TLB entry is the thing that contains this translation



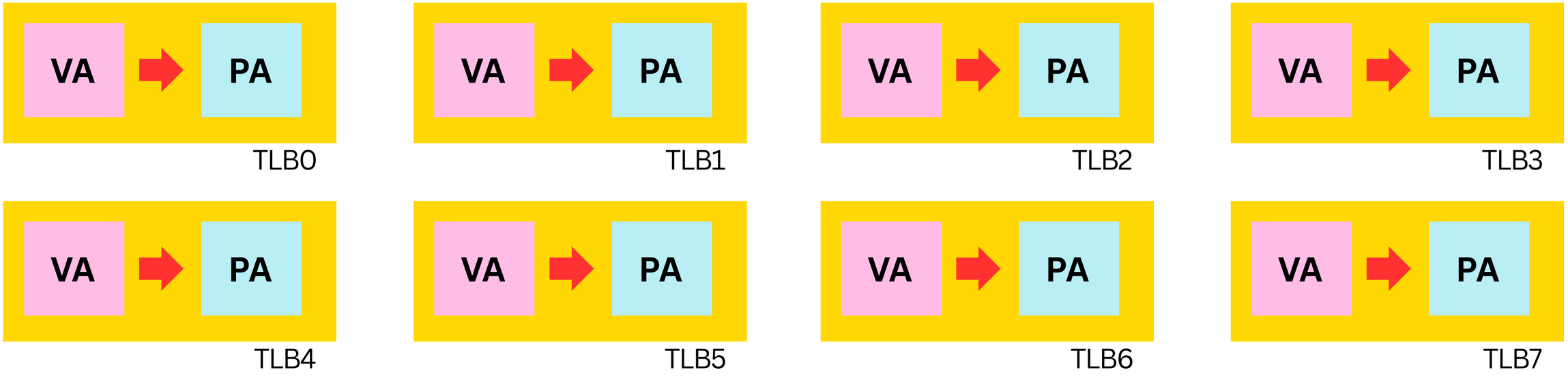
We give it a virtual address, a physical address (and some other information), and then the computer stores it somewhere on the coprocessor



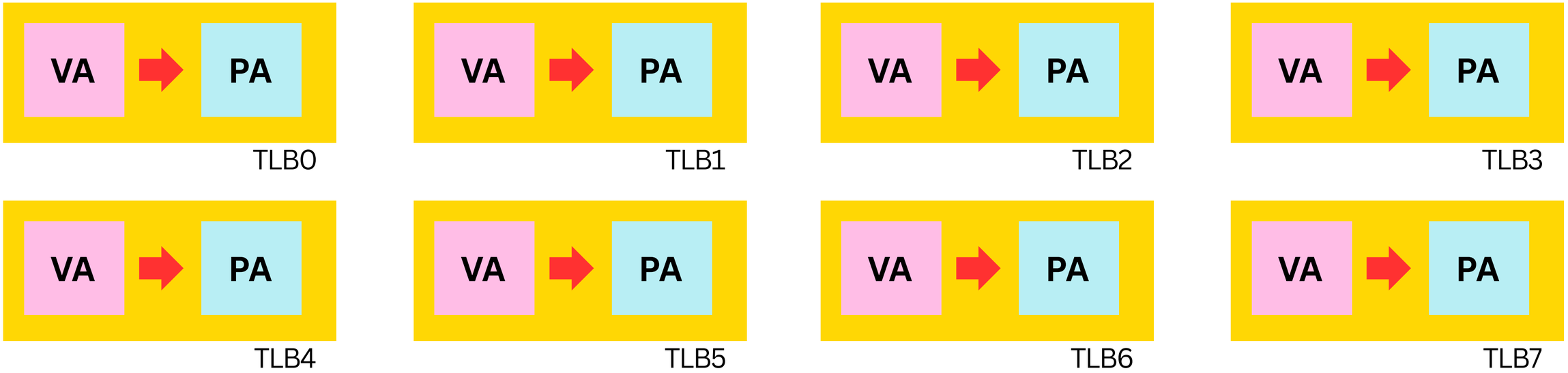
The co-processor has space to hold 8 such mappings



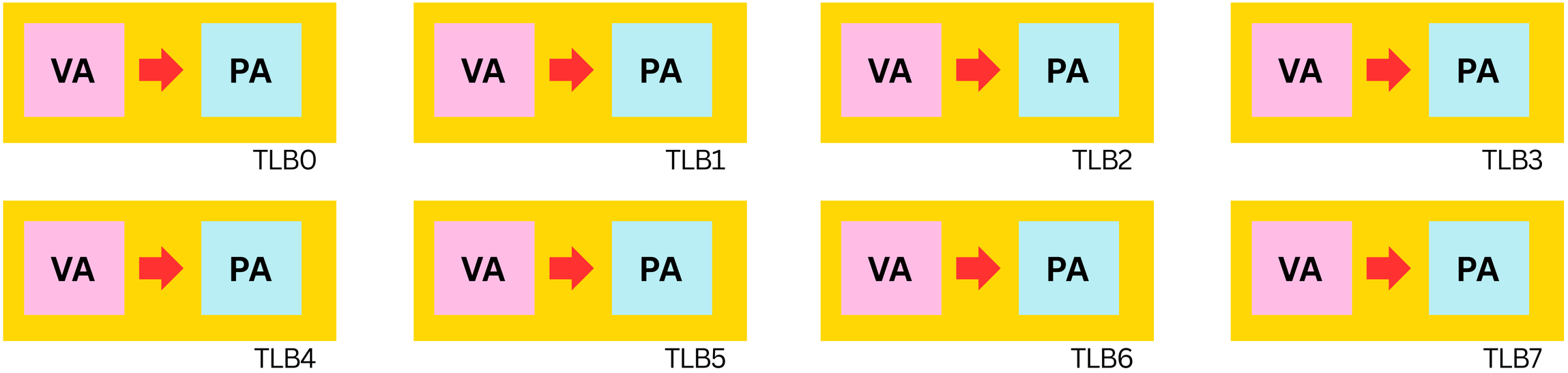
You might see a problem here...



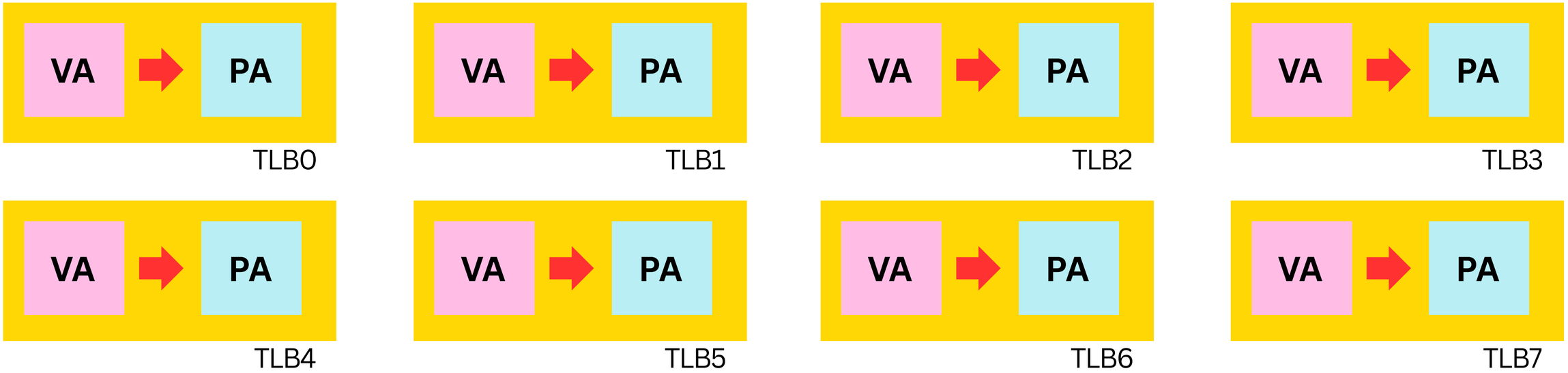
In real life we probably have a whole bunch of addresses we wanna translate



8 is probably not enough...



We will return to this problem later, but first back to the TLB



Remember that each TLB entry requires the use of these 4 registers

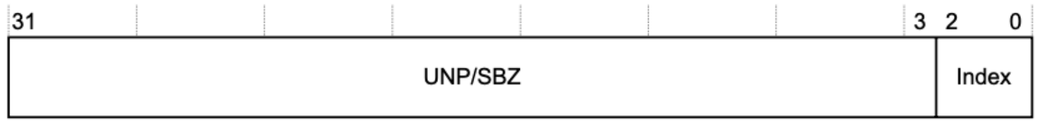


Figure 3-76 TLB Lockdown Index Register format

Figure 3-77 TLB Lockdown VA Register format

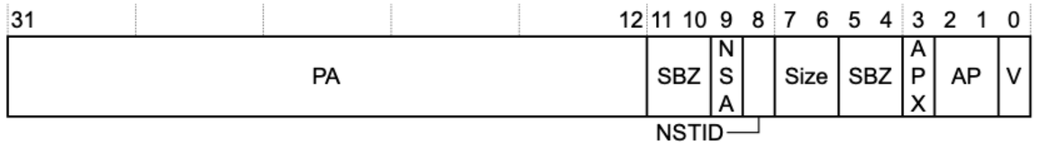


Figure 3-78 TLB Lockdown PA Register format

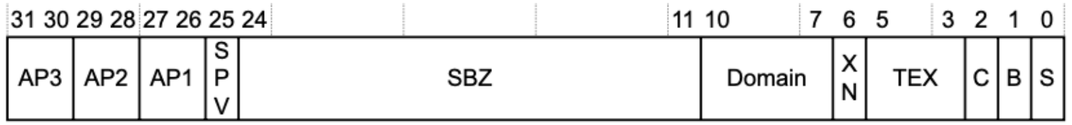
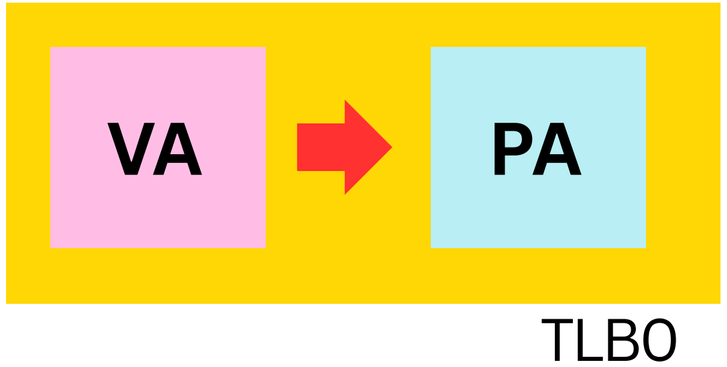


Figure 3-79 TLB Lockdown Attributes Register format



So you can think of these 4 registers coming together to make TLB0

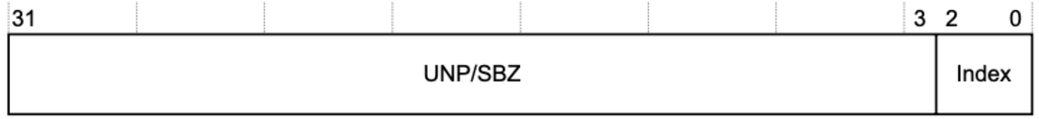


Figure 3-76 TLB Lockdown Index Register format

Figure 3-77 TLB Lockdown VA Register format

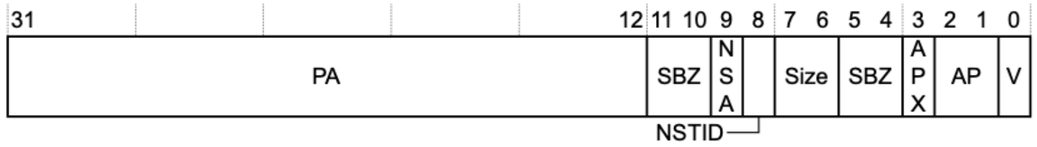


Figure 3-78 TLB Lockdown PA Register format

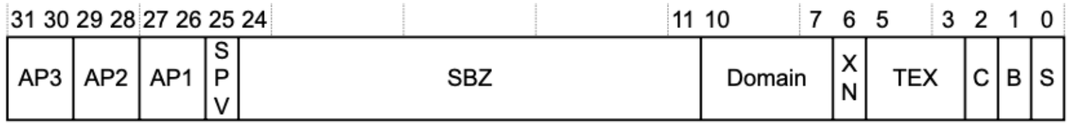
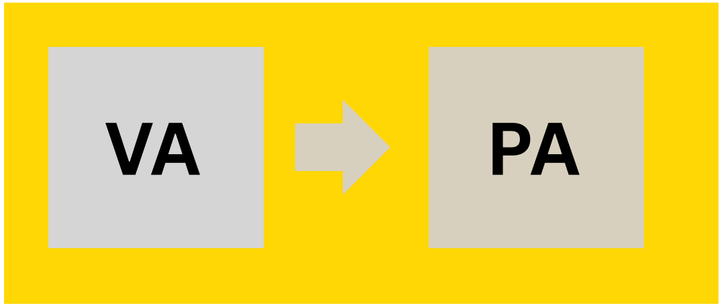


Figure 3-79 TLB Lockdown Attributes Register format



TLB0

We put the Virtual address here

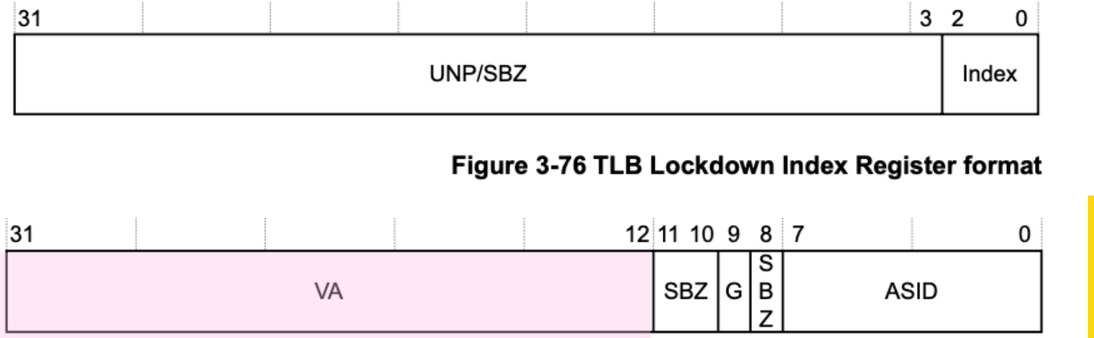


Figure 3-76 TLB Lockdown Index Register format

Figure 3-77 TLB Lockdown VA Register format

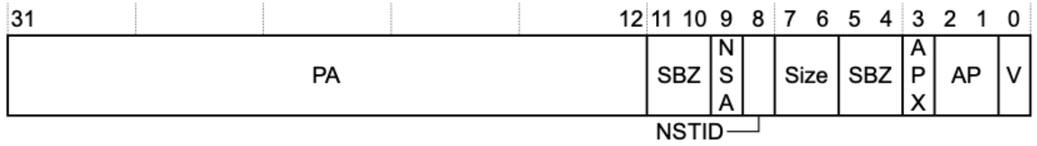
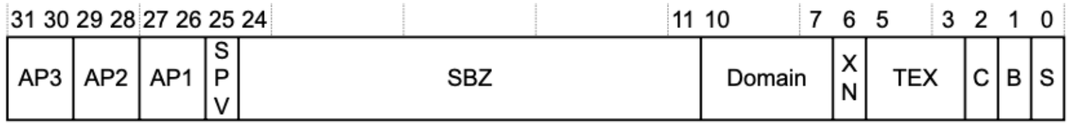


Figure 3-78 TLB Lockdown PA Register format



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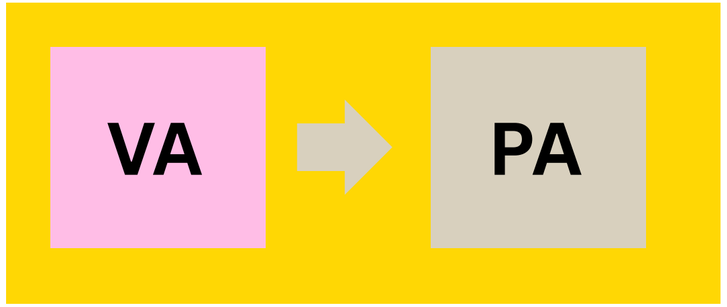


TLB0

And the phycial address here

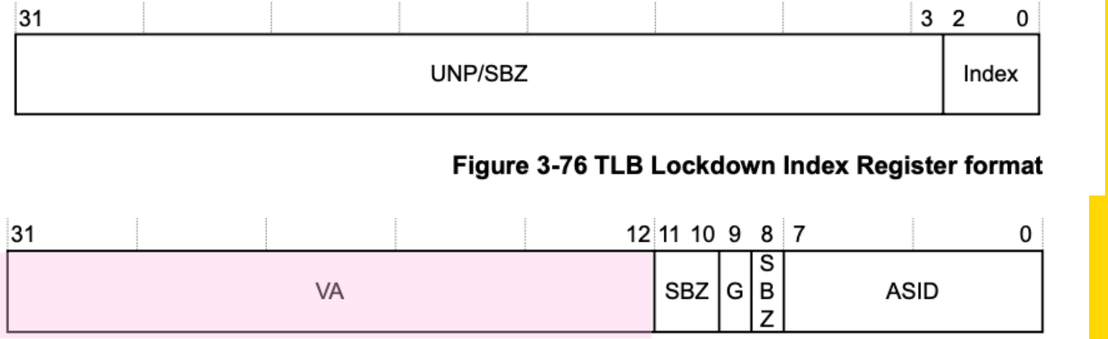
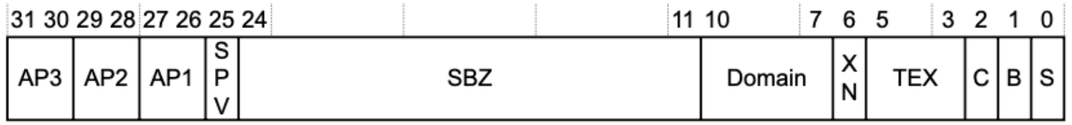


Figure 3-76 TLB Lockdown Index Register format

Figure 3-77 TLB Lockdown VA Register format

Figure 3-78 TLB Lockdown PA Register format



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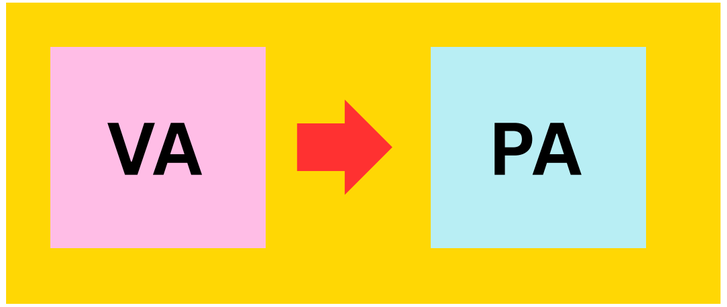


TLB0

The other important things are ASID

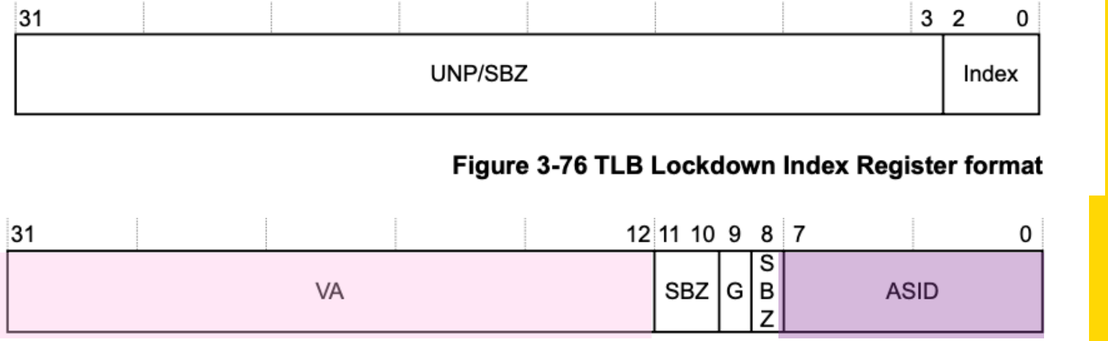


Figure 3-76 TLB Lockdown Index Register format

Figure 3-77 TLB Lockdown VA Register format

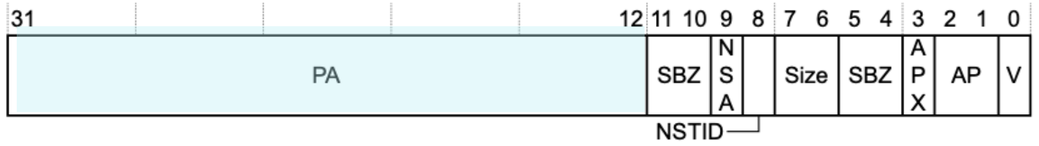


Figure 3-78 TLB Lockdown PA Register format

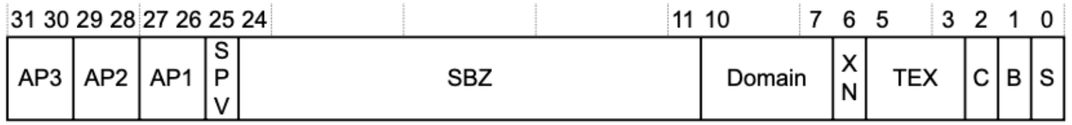
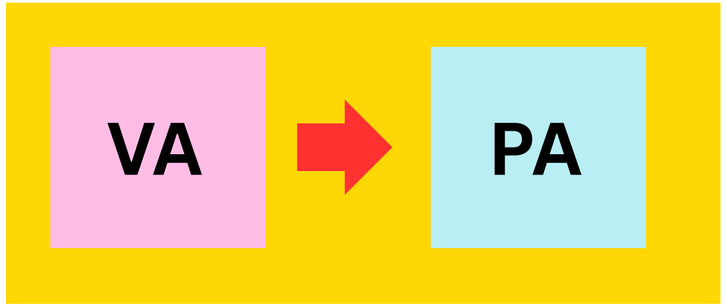


Figure 3-79 TLB Lockdown Attributes Register format



TLB0

And size

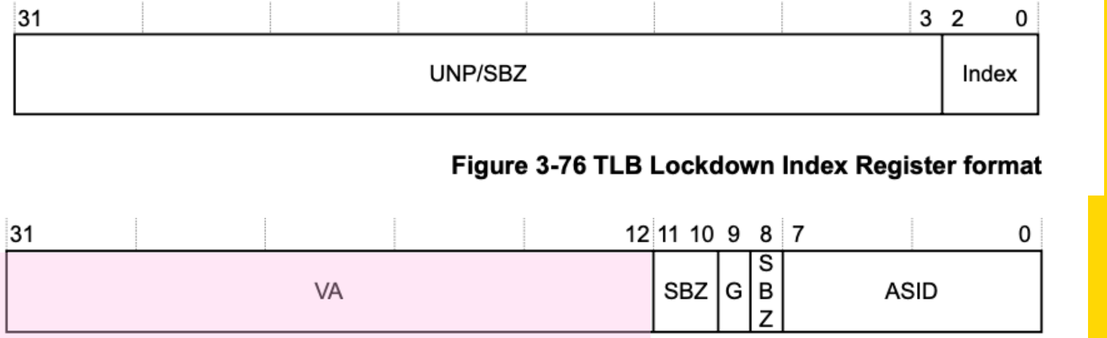


Figure 3-76 TLB Lockdown Index Register format

Figure 3-77 TLB Lockdown VA Register format

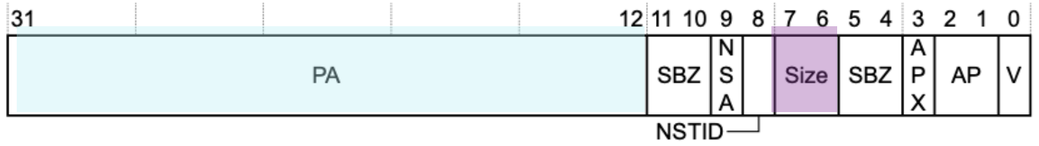


Figure 3-78 TLB Lockdown PA Register format

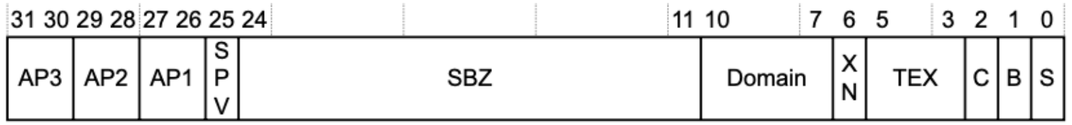
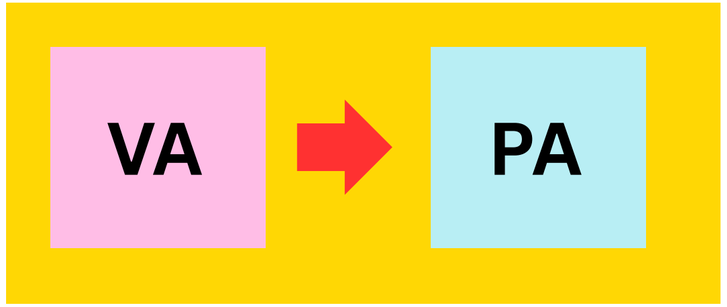


Figure 3-79 TLB Lockdown Attributes Register format



TLB0

Size is easy...

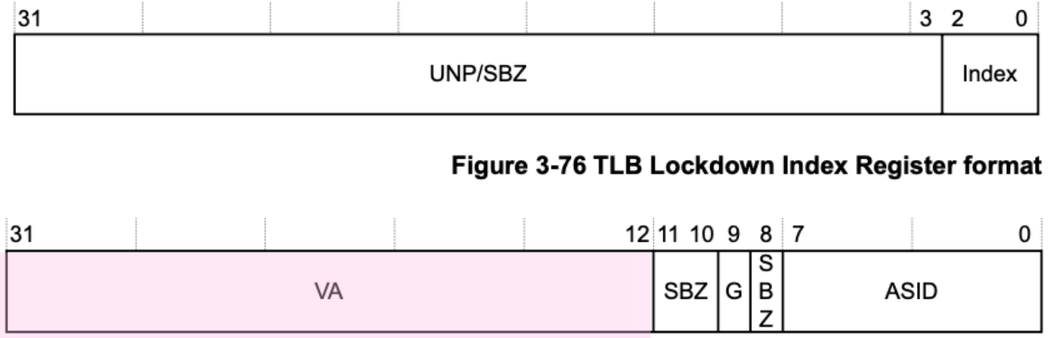


Figure 3-76 TLB Lockdown Index Register format

Figure 3-77 TLB Lockdown VA Register format

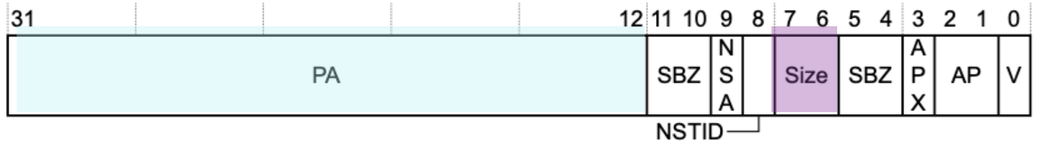


Figure 3-78 TLB Lockdown PA Register format

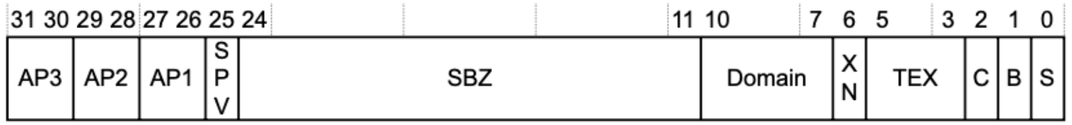
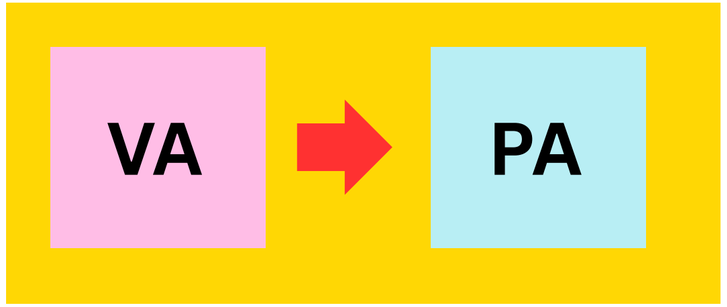


Figure 3-79 TLB Lockdown Attributes Register format



TLB0

It just tells you how much memory from the start address you want to map

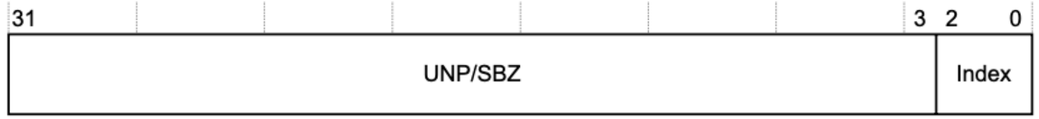


Figure 3-76 TLB Lockdown Index Register format

Figure 3-77 TLB Lockdown VA Register format

Figure 3-78 TLB Lockdown PA Register format

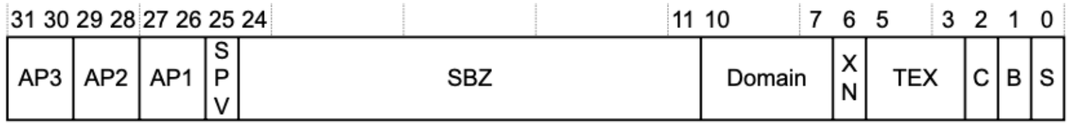


Figure 3-79 TLB Lockdown Attributes Register format

Defines the size of the memory region that this page table entry describes:

b00 = 16MB supersection

b01 = 4KB page

b10 = 64KB page

b11 = 1M section.

ASID is more tricky, but important to understand

Figure 3-76 TLB Lockdown Index Register format

Figure 3-77 TLB Lockdown VA Register format

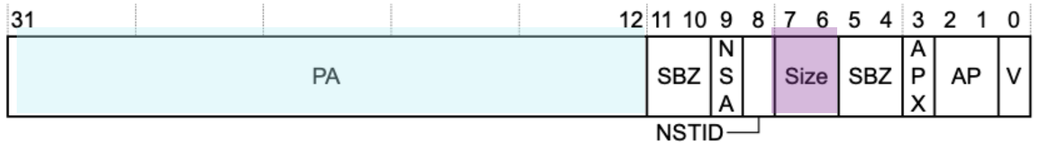


Figure 3-78 TLB Lockdown PA Register format

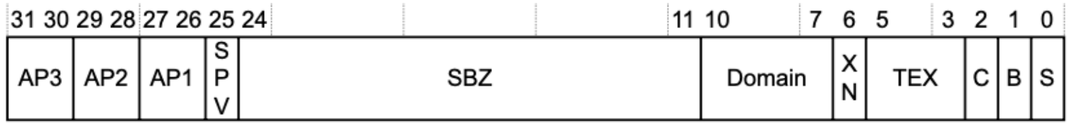
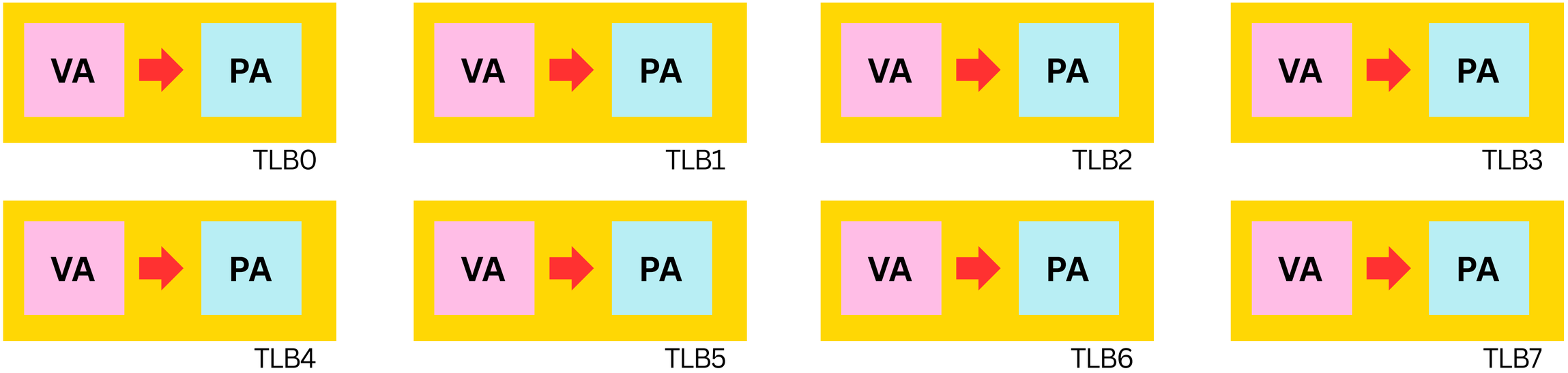


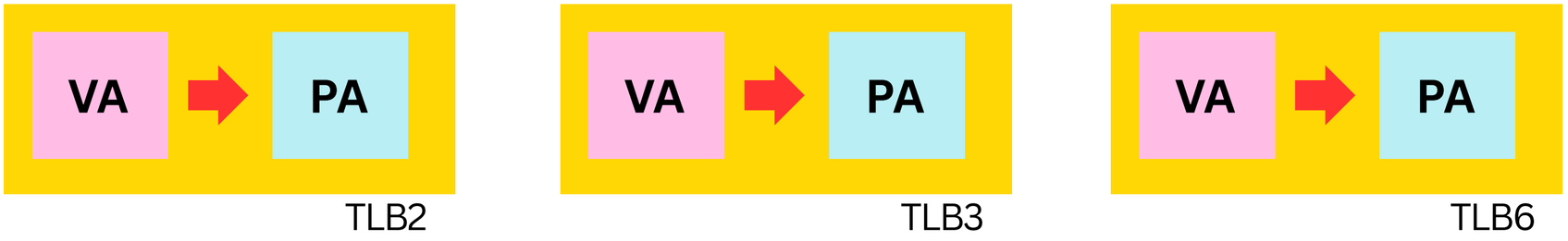
Figure 3-79 TLB Lockdown Attributes Register format

?

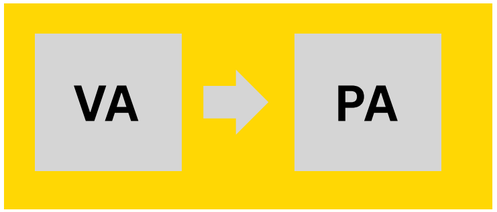
Lets come back here to understand why



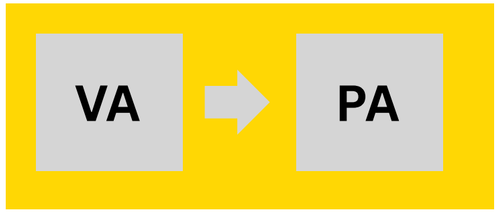
The point of VM is to make it seem like each thread has the same Virtual address, but different physical Address



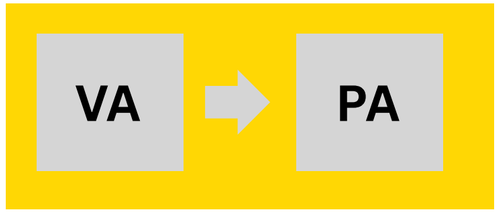
Lets say we make 3 threads



Thread 1

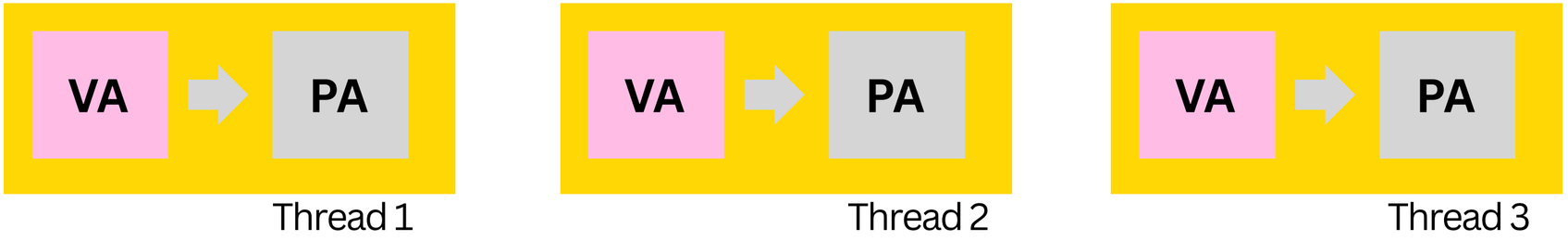


Thread 2

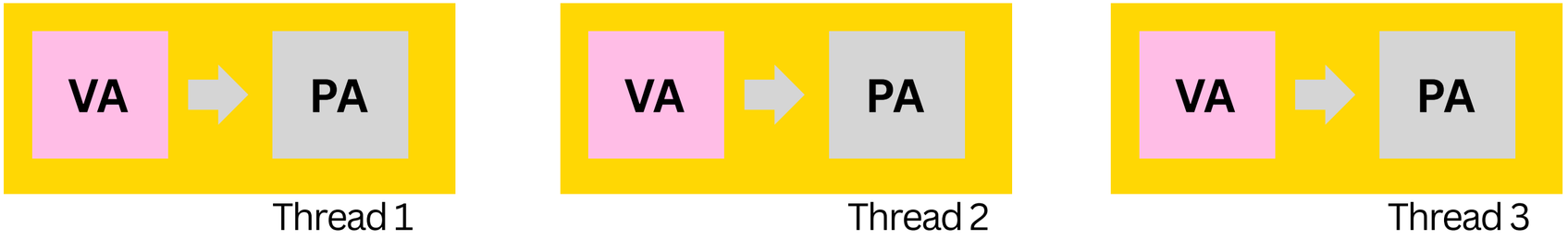


Thread 3

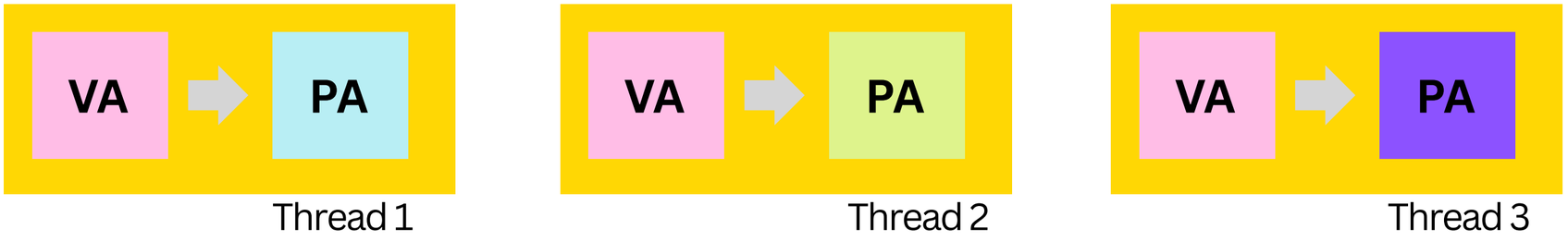
We give them all the same Virtual address



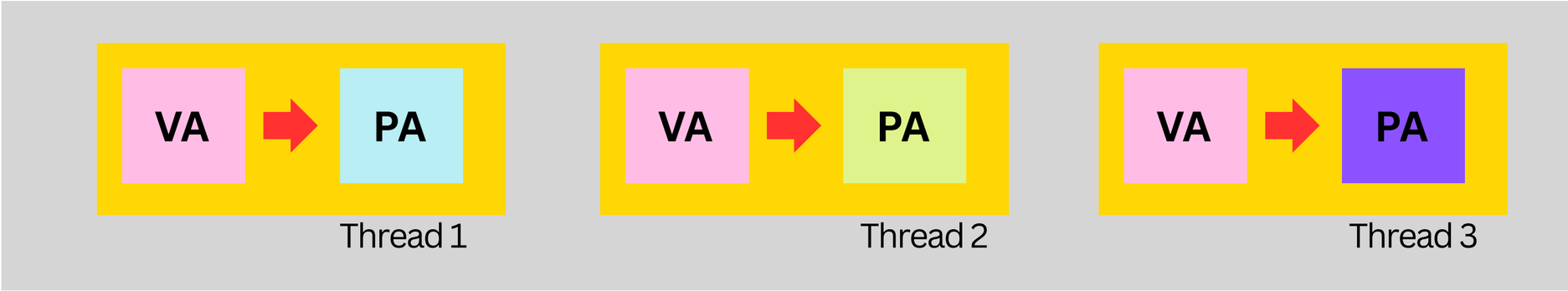
Now we have to assign them different physical addresses

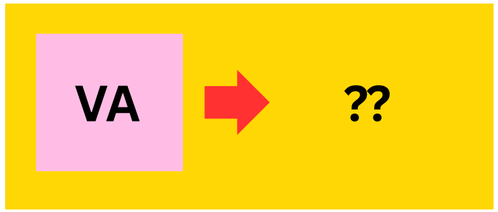


These are the different addresses

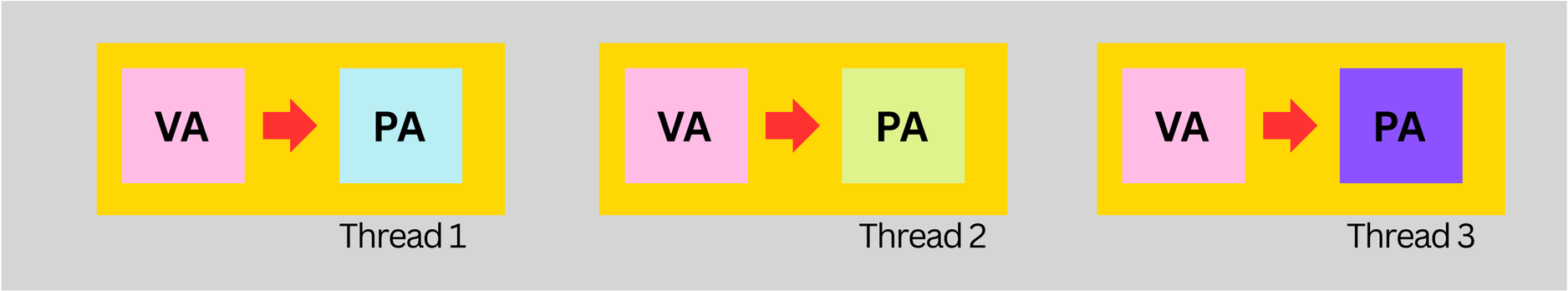


We put it in the TLB

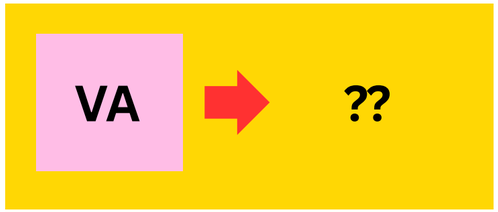




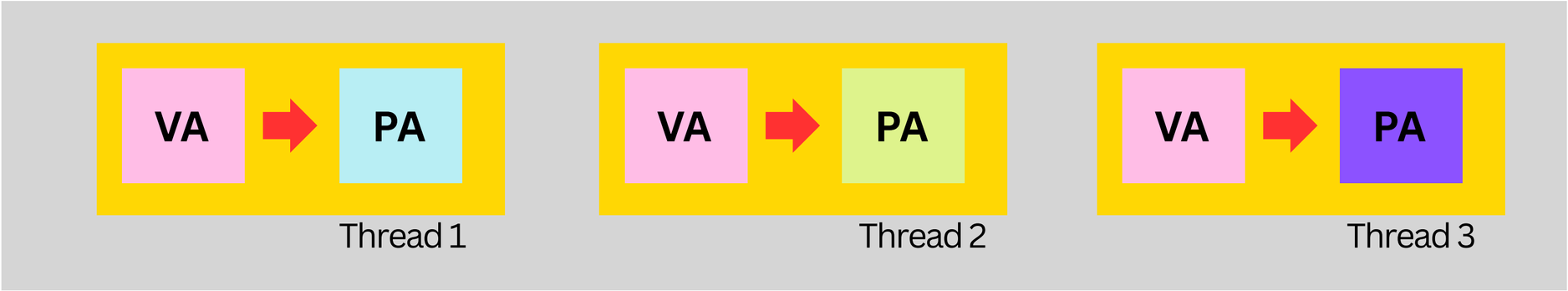
We put it in the TLB



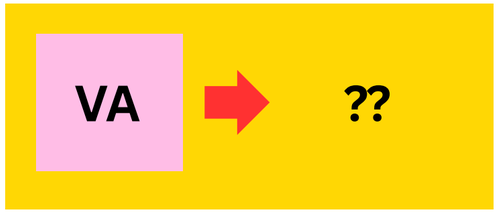
Now Thread 1 is running, and asks for its Virtual Address to be converted



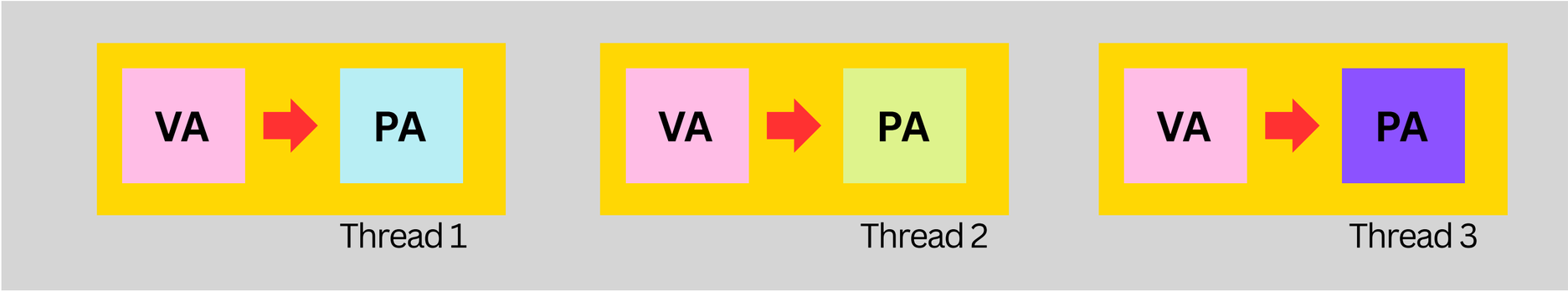
We put it in the TLB



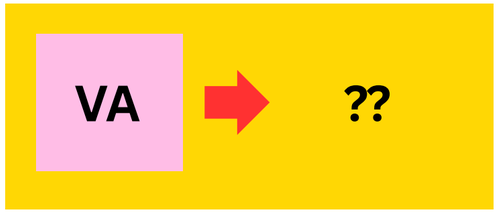
The CPU looks in the TLB. There are three identical VAs that map to different PAs



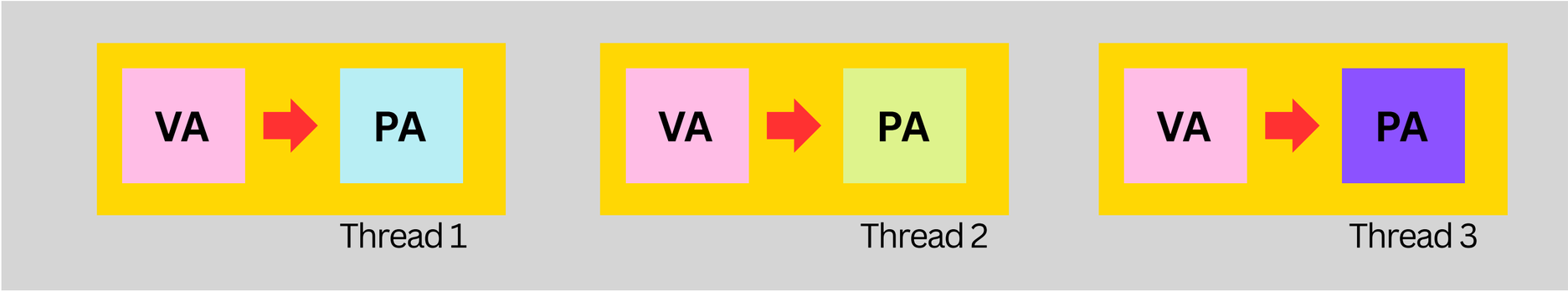
We put it in the TLB



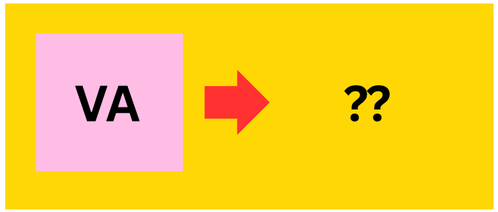
How does it know what address to return?



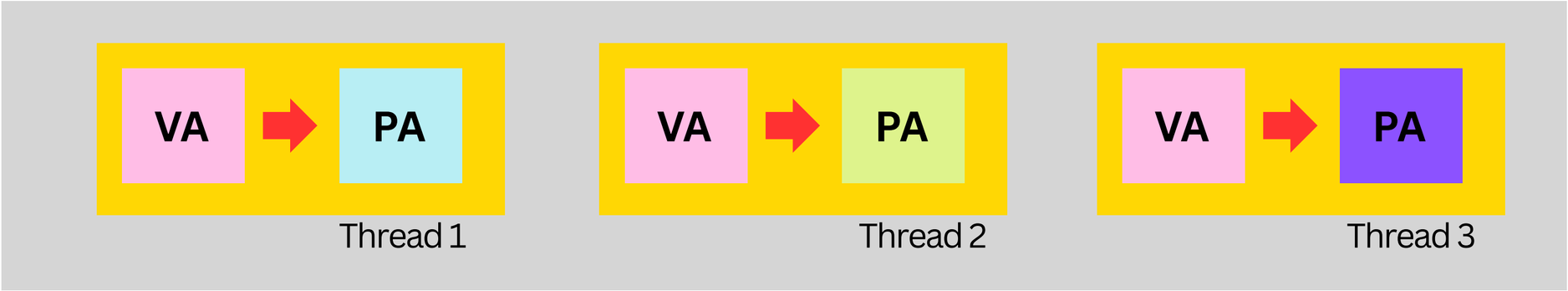
We put it in the TLB



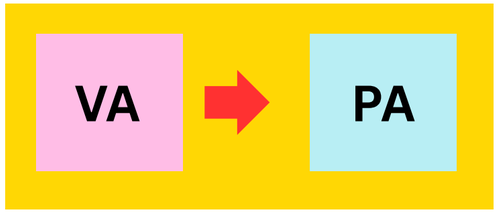
It needs to know which thread is running ...



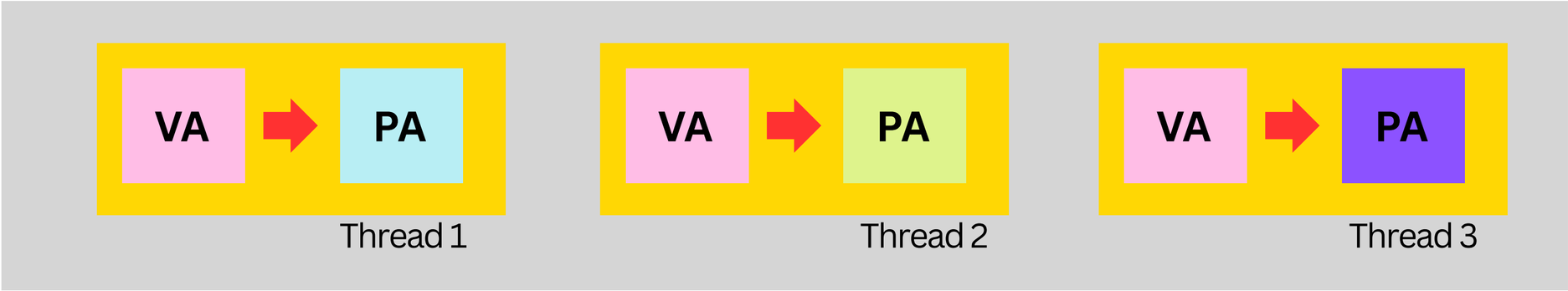
We put it in the TLB



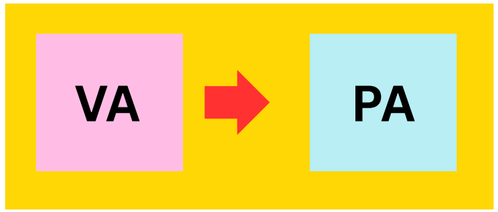
If it knows Thread 1 is running, then it knows which PA to give



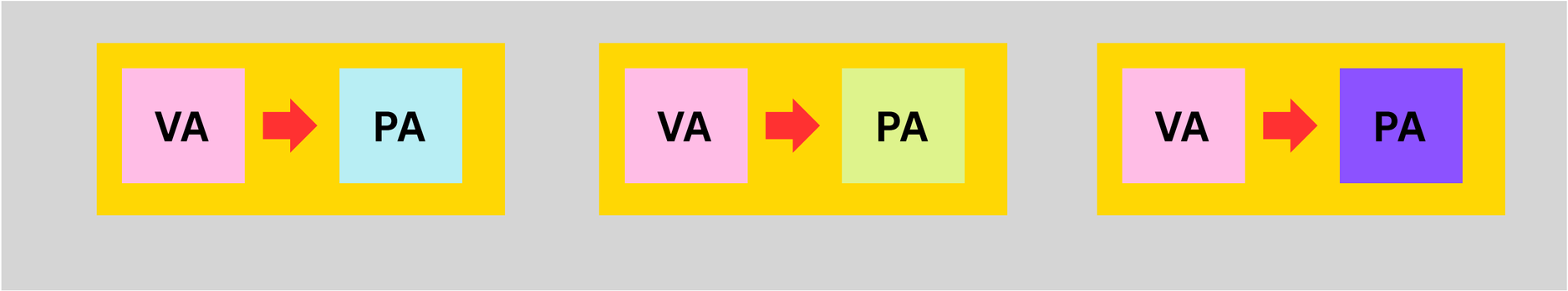
We put it in the TLB



It would be helpful to have another feild here

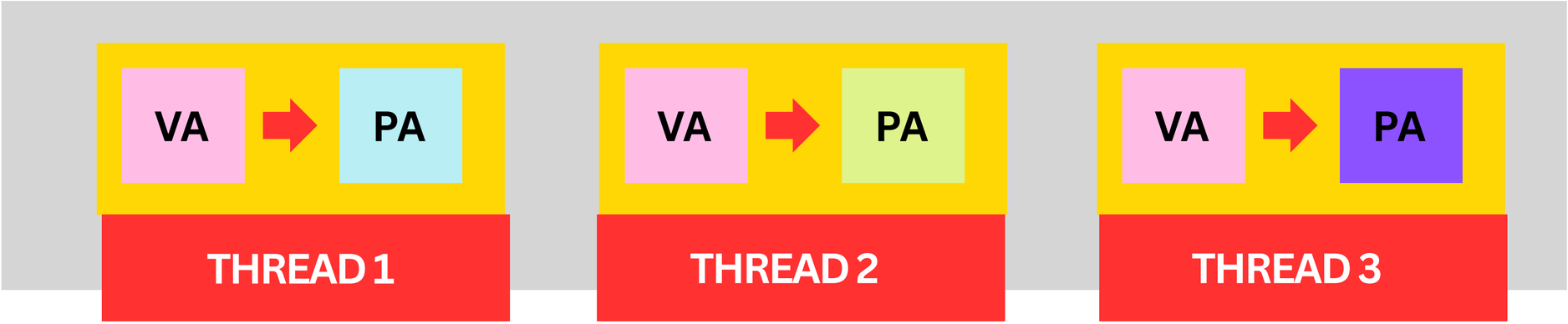


We put it in the TLB

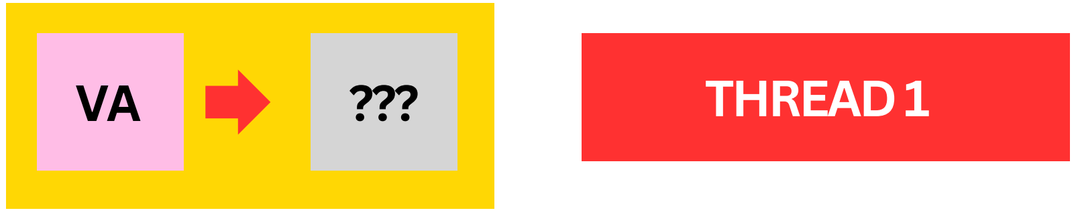


Something like who is running?

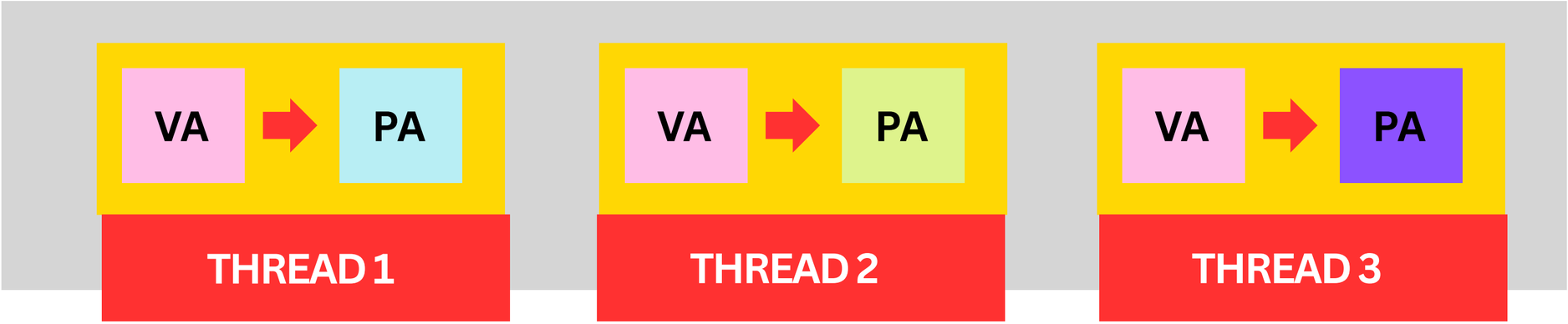
We put it in the TLB



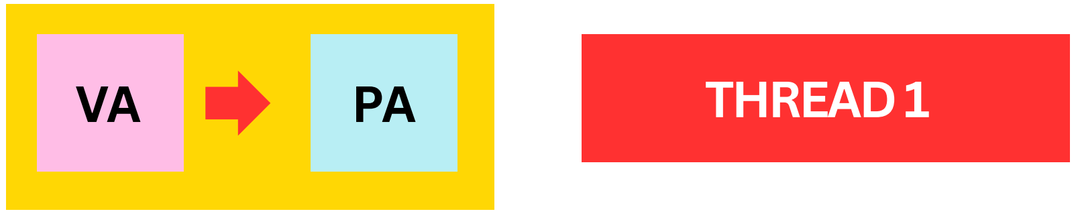
Something that helps us map who is asking



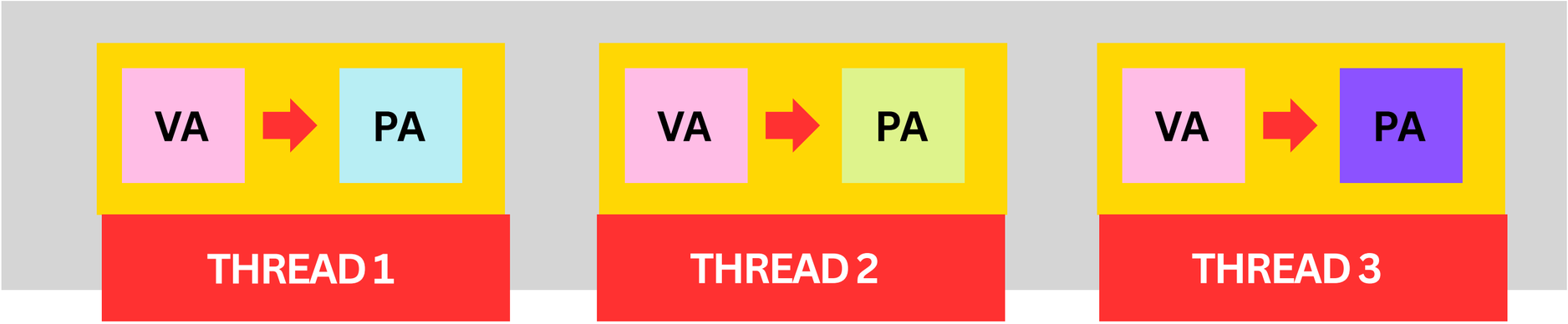
We put it in the TLB



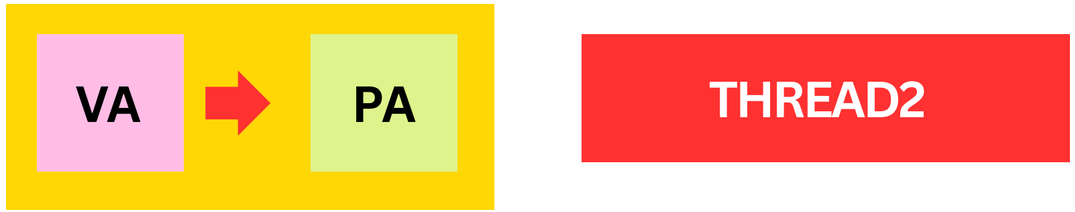
Converting is easy, when we know who is asking, and what to give depending on that



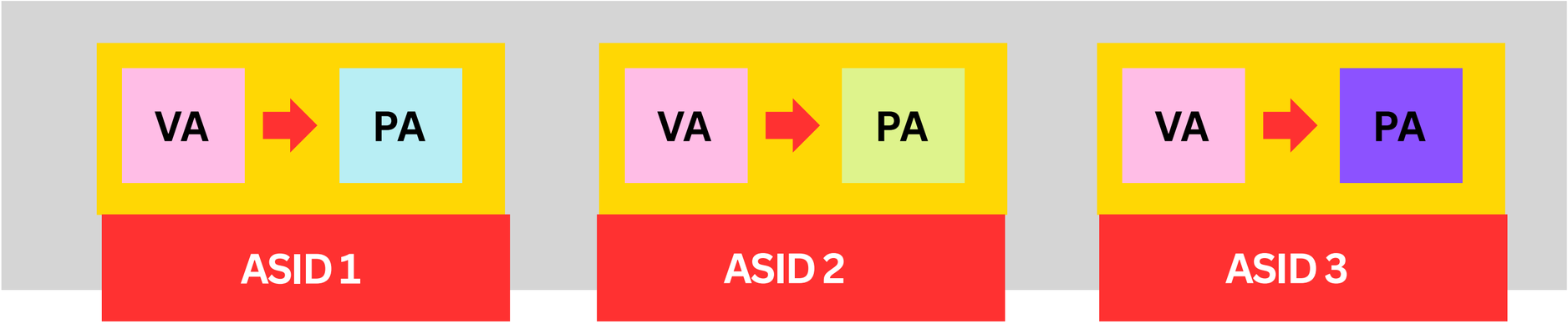
We put it in the TLB



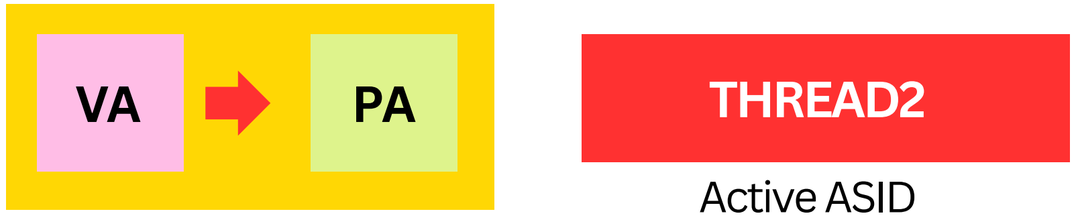
So if Thread 2 was asking we can give its' PA



We can all this feild the thread's ASID



Given which ASID IS active, and which VA we want to convert, we can get the PA



So each process is given an ASID

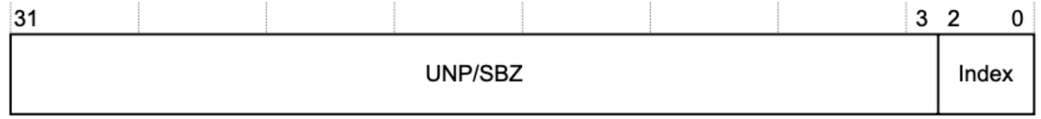


Figure 3-76 TLB Lockdown Index Register format

Figure 3-77 TLB Lockdown VA Register format

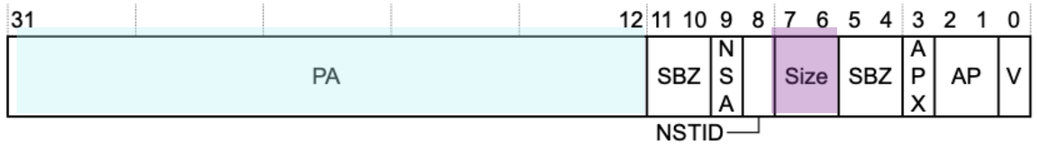


Figure 3-78 TLB Lockdown PA Register format

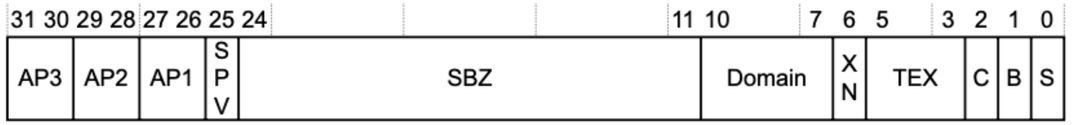


Figure 3-79 TLB Lockdown Attributes Register format

ASID =
Address Space ID

There’s one more register you have to be aware of

Figure 3-64 DMA Context ID Register format

There is only one of these

Figure 3-64 DMA Context ID Register format

It holds the ASID of the thread/ process thats currently active

Figure 3-64 DMA Context ID Register format

And the process ID, which we don't really use in this class

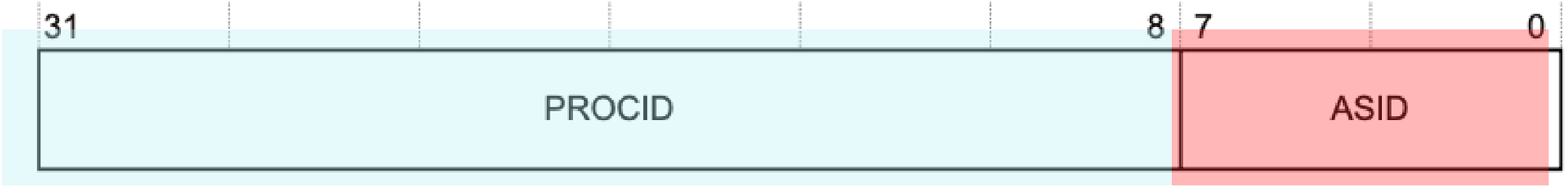


Figure 3-64 DMA Context ID Register format

So it tells us which process is running/ asking for the VA translation

Figure 3-64 DMA Context ID Register format

Whenever you switch into a thread, you need to update this register

Figure 3-64 DMA Context ID Register format

Whenever you switch into a thread, you need to update this register

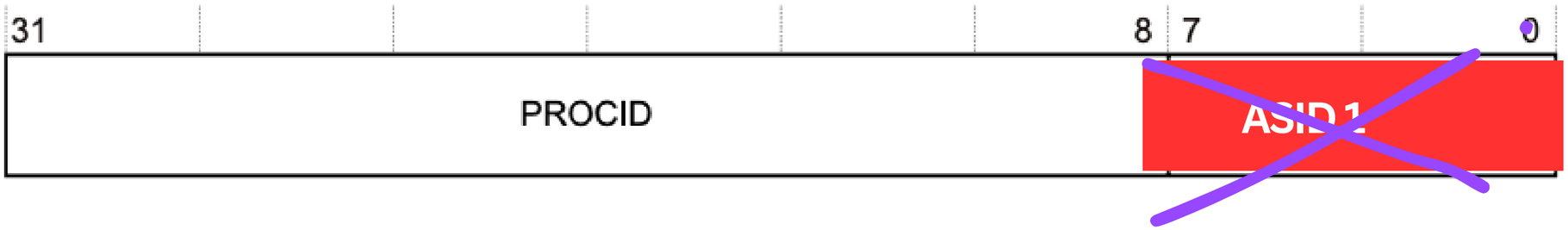


Figure 3-64 DMA Context ID Register format

Whenever you switch into a thread, you need to update this register

Figure 3-64 DMA Context ID Register format

So that when the thread asks for a converstion, you can tell the processor “who’s asking!”

Figure 3-64 DMA Context ID Register format

The processor will first check the ASID section in the TLB

Figure 3-64 DMA Context ID Register format

Then will check which PA the VA is mapped to

Figure 3-64 DMA Context ID Register format

And that’s all you need to understand for the first part of this lab!

Questions you should be able to answer

- What does the TLB do?
- Theoretically, whats the maximum number of Threads that the TLB can support?
- Why is the ASID required?